

Polarization of Gravitational Waves From Turbulence: Tension Between Simulation and Analytic Approaches

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I. GRAVITATIONAL WAVES FROM A TURBULENT SOURCE

Before specializing to particular spectral and temporal models, we collect the standard analytic derivation of the gravitational-wave (GW) spectrum sourced by turbulence [1–5]. This fixes the conventions used throughout and, in particular, shows why the remaining sections reduce the problem to computing the contracted source correlator $H_{ijij}(\mathbf{k}, \omega)$ of Eq. (10) on the diagonal $\omega = k$.

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A. Sourced wave equation

Write the tensor perturbations of a spatially flat FLRW background as

$$ds^2 = a^2(\tau)[-d\tau^2 + (\delta_{ij} + h_{ij})dx^i dx^j], \quad \partial_i h_{ij} = h_{ii} = 0, \quad (1)$$

with h_{ij} transverse-traceless (TT). The linearized Einstein equations give

$$h''_{ij} + 2\frac{a'}{a}h'_{ij} - \nabla^2 h_{ij} = 16\pi G a^2 \Pi_{ij}^{\text{TT}}, \quad (2)$$

where $' = d/d\tau$ and Π_{ij}^{TT} is the TT part of the anisotropic stress. For an incompressible, non-relativistic turbulent fluid the source is the Reynolds stress,

$$\Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau) = \Lambda_{ij,lm}(\hat{k}) T_{lm}(\mathbf{k}, \tau), \quad T_{lm} = w v_l v_m, \quad \Lambda_{ij,lm} = P_{il}P_{jm} - \frac{1}{2}P_{ij}P_{lm}, \quad (3)$$

with $w = \bar{\rho} + \bar{p}$ the enthalpy density, $P_{ij} = \delta_{ij} - \hat{k}_i \hat{k}_j$, and Λ the TT projector. The double TT projection together with the Gaussian (Wick) reduction of the quartic velocity correlator collapses the two-point function of (3) onto the contracted form $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ that appears in Eq. (10).

Turbulence acts on eddy-turnover times short compared with a Hubble time, so over the source lifetime we drop the expansion ($a \approx 1$, friction negligible) and Eq. (2) becomes, in Fourier space, a forced oscillator,

$$h''_{ij}(\mathbf{k}, \tau) + k^2 h_{ij}(\mathbf{k}, \tau) = 16\pi G \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau). \quad (4)$$

B. Green-function solution and the GW spectrum

The retarded solution of Eq. (4) and its time derivative are

$$h_{ij}(\mathbf{k}, \tau) = \frac{16\pi G}{k} \int_{\tau_i}^{\tau} d\tau' \sin[k(\tau - \tau')] \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau'), \quad \dot{h}_{ij}(\mathbf{k}, \tau) = 16\pi G \int_{\tau_i}^{\tau} d\tau' \cos[k(\tau - \tau')] \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau'). \quad (5)$$

With the Isaacson energy density $\rho_{\text{GW}} = \langle \dot{h}_{ij} \dot{h}_{ij} \rangle / (32\pi G)$ and the unequal-time correlator (UETC) of the TT stress,

$$\langle \Pi_{ij}^{\text{TT}}(\mathbf{k}, \tau_1) \Pi_{ij}^{\text{TT}*}(\mathbf{k}', \tau_2) \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}') \Pi(k; \tau_1, \tau_2), \quad (6)$$

inserting (5) and averaging the product $\cos[k(\tau - \tau_1)] \cos[k(\tau - \tau_2)] \rightarrow \frac{1}{2} \cos[k(\tau_1 - \tau_2)]$ over the fast GW oscillations gives the master formula

$$\frac{d\rho_{\text{GW}}}{d \ln k} \simeq \frac{2G k^3}{\pi} \int d\tau_1 \int d\tau_2 \cos[k(\tau_1 - \tau_2)] \Pi(k; \tau_1, \tau_2). \quad (7)$$

The GW spectrum is thus the temporal Fourier transform of the source UETC, evaluated at the on-shell frequency $\omega = k$. For a statistically stationary (or quasi-stationary) source, $\Pi(k; \tau_1, \tau_2) = \Pi(k; \tau_1 - \tau_2)$ and the double time integral factorizes into the source duration T_{src} times the temporal transform $\tilde{\Pi}(k, \omega) = \int d(\Delta\tau) e^{i\omega\Delta\tau} \Pi(k; \Delta\tau)$, which is precisely the spatio-temporal correlator $H_{ijij}(\mathbf{k}, \omega)$ of Eq. (10). Hence

$$\frac{d\rho_{\text{GW}}}{d \ln k} \propto k^3 H_{ijij}(\mathbf{k}, \omega)|_{\omega=k} \implies \Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \quad (8)$$

the GW dispersion $\omega = k$ becoming the diagonal $q = p$ used in Sec. VI (with characteristic strain $h_c^2(p) = p H(p, p)$). Everything downstream is therefore an evaluation of H_{ijij} , whose spatial part is a convolution of the velocity power spectrum (quadratic source, Eqs. (11)–(12)) and whose temporal part is the decorrelation model carried by $\Pi(k; \tau_1, \tau_2)$.

C. Spectral shape from the literature

For Kolmogorov inertial-range input the analytic results have three robust pieces:

- *Infrared* ($k \ll k_0$): causality forces the stress correlator to be analytic (white) as $k \rightarrow 0$, so $H \rightarrow \text{const}$ and $\Omega_{\text{GW}} \propto k^3$ [3, 4].
- *Peak* ($k \sim k_0$): set by the largest eddies, with parametric amplitude $(d\Omega_{\text{GW}}/d\ln k)_{\text{peak}} \sim (H_*/k_0) \Omega_K^2$, i.e. the squared kinetic-energy fraction suppressed by the source-duration factor $H_*/k_0 \sim H_*\tau_{\text{eddy}}$ [1, 3].
- *Ultraviolet* ($k \gg k_0$): *model-dependent*, fixed by the temporal decorrelation $\Pi(k; \tau_1, \tau_2)$. Kraichnan sweeping gives the Gogoberidze tail [2], while a δ -correlated or a decaying-turbulence kernel give different slopes.

The UV slope is precisely where the four UETC models derived below (δ -spectrum vs. white-noise, Kraichnan vs. decay) differ, which is the focus of the remainder of the paper.

II. KRAICHNAN-K41 MODEL (GOGOBERIDZE 2007)

We use the spatial/temporal Fourier pair $R_{ij}(\mathbf{r}, \tau) = \int \frac{d^3k}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{r}} F_{ij}(\mathbf{k}, \tau)$ and $f(\eta_k, \tau) = \int \frac{d\omega}{2\pi} e^{-i\omega\tau} \tilde{f}(\eta_k, \omega)$, and adopt the isotropic incompressible model of [2] (Eqs. 23–26),

$$F_{ij}(\mathbf{k}, \tau) = A(k) P_{ij}(\hat{k}) f(\eta_k, \tau), \quad P_{ij}(\hat{k}) = \delta_{ij} - \hat{k}_i \hat{k}_j, \quad (9)$$

with $A(k) = C_k \epsilon^{2/3} k^{-11/3} / 4\pi$, $f(\eta_k, \tau) = \exp(-\frac{\pi}{4} \eta_k^2 \tau^2)$, and $\eta_k = \frac{1}{\sqrt{2\pi}} \epsilon^{1/3} k^{2/3}$.

The fourth-rank Fourier-space correlator $H_{ijkl}(\mathbf{k}, \omega) = (2\pi)^{-4} \int d\tau d^3\xi e^{i\omega\tau - i\mathbf{k}\cdot\xi} R_{ijkl}(\xi, \tau)$ with the contraction $R_{ijij} = R_{ii}R_{jj} + \frac{1}{3}R_{ij}R_{ij}$ (Gogoberidze Eq. 28) reads, in real space,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^4} \int d\tau d^3\mathbf{r} e^{i\omega\tau - i\mathbf{k}\cdot\mathbf{r}} \left[R_{ii}(\mathbf{r}, \tau) R_{jj}(\mathbf{r}, \tau) + \frac{1}{3} R_{ij}(\mathbf{r}, \tau) R_{ij}(\mathbf{r}, \tau) \right]. \quad (10)$$

Inserting the spatial Fourier representation of R_{ij} , the $\mathbf{r}$ integral produces a momentum-conserving $\delta^{(3)}(\mathbf{k} - \mathbf{q} - \mathbf{p})$. Renaming $\mathbf{q} \rightarrow \mathbf{k}_1$, $\mathbf{u} = \mathbf{k} - \mathbf{k}_1$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int d\tau e^{i\omega\tau} d^3k_1 \left[F_{ii}(\mathbf{k}_1, \tau) F_{jj}(\mathbf{u}, \tau) + \frac{1}{3} F_{ij}(\mathbf{k}_1, \tau) F_{ij}(\mathbf{u}, \tau) \right]. \quad (11)$$

The temporal Fourier transform then converts the τ integral into a frequency δ -function, leaving the frequency convolution

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 d\omega_1 \left[\tilde{F}_{ii}(\mathbf{k}_1, \omega_1) \tilde{F}_{jj}(\mathbf{u}, \omega - \omega_1) + \frac{1}{3} \tilde{F}_{ij}(\mathbf{k}_1, \omega_1) \tilde{F}_{ij}(\mathbf{u}, \omega - \omega_1) \right]. \quad (12)$$

Substituting Eq. (9) and using $P_{ii}(\hat{k}) = 2$ together with $P_{ij}(\hat{k}_1) P_{ij}(\hat{u}) = 1 + (\hat{k}_1 \cdot \hat{u})^2$, the tensor contractions reduce Eq. (12) to

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^8} \int d^3k_1 d\omega_1 A(k_1) A(u) \tilde{f}(\eta_{k_1}, \omega_1) \tilde{f}(\eta_u, \omega - \omega_1) \left[4 + \frac{1}{3} (1 + (\hat{k}_1 \cdot \hat{u})^2) \right]. \quad (13)$$

This is the Fourier representation valid for all k ; we treat $\mathbf{k} > \mathbf{k}_0$ and $\mathbf{k} \leq \mathbf{k}_0$ separately below.

A. $\mathbf{k} > \mathbf{k}_0$

The law of cosines $\hat{k}_1 \cdot \hat{u} = (k^2 - k_1^2 - u^2)/(2k_1u)$ turns the bracket in Eq. (13) into a homogeneous polynomial in (k, k_1, u) ; expanding and grouping gives the geometric kernel

$$\mathcal{K}(k, k_1, u) \equiv \frac{27}{6} + \frac{k^4}{12k_1^2u^2} + \frac{k_1^2}{12u^2} + \frac{u^2}{12k_1^2} - \frac{k^2}{6u^2} - \frac{k^2}{6k_1^2} \quad (\text{cf. Eq. 29 of [2]}). \quad (14)$$

The temporal transform $\tilde{f}(\eta_k, \omega) = (2/\eta_k) \exp(-\omega^2/\pi\eta_k^2)$ combines with $A(k)$ into the single spectral factor

$$g(k, \omega) \equiv A(k) \tilde{f}(\eta_k, \omega) = \frac{2E_k}{4\pi k^2 \eta_k} \exp\left(-\frac{\omega^2}{\pi\eta_k^2}\right), \quad E_k = C_k \epsilon^{2/3} k^{-5/3}. \quad (15)$$

Going to spherical coordinates for $\mathbf{k}_1$ ($d^3k_1 = k_1^2 dk_1 d\phi d\mu$ with $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}_1$), the azimuthal integral contributes 2π . Trading μ for $u = \sqrt{k^2 + k_1^2 - 2kk_1\mu}$ via $d\mu = u du/(kk_1)$, with $u \in [|k - k_1|, k + k_1]$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{1}{(2\pi)^7} \int_0^\infty \frac{k_1 dk_1}{k} \int_{|k-k_1|}^{k+k_1} u du \int_{-\infty}^\infty d\omega_1 g(k_1, \omega_1) g(u, \omega - \omega_1) \mathcal{K}(k, k_1, u). \quad (16)$$

Switching to the dimensionless variables $q = k_1/k$, $s = u/k$, $\Omega = \omega/(\epsilon^{1/3}k^{2/3})$, $\Omega_1 = \omega_1/(\epsilon^{1/3}k^{2/3})$, g becomes $g(k, \omega) = (1/\sqrt{2\pi}) C_k \epsilon^{1/3} k^{-13/3} \exp(-2\Omega^2)$. The product $g(k_1, \omega_1)g(u, \omega - \omega_1)$ thus separates into a dimensional prefactor $\propto k^{-26/3}$ times a manifestly dimensionless Gaussian envelope $\Phi(q, \Omega_1; s, \Omega - \Omega_1) = q^{-13/3} s^{-13/3} \exp[-2\Omega_1^2 q^{-4/3} - 2(\Omega - \Omega_1)^2 s^{-4/3}]$. Collecting all k -dependent factors gives the scaling form

$$H_{ijij}(\mathbf{k}, \omega) = \frac{C_k^2 \epsilon}{256 \pi^8} k^{-5} \mathfrak{H}(\Omega), \quad \mathfrak{H}(\Omega) = \int_0^\infty dq q \int_{-\infty}^\infty d\Omega_1 \int_{|1-q|}^{1+q} ds s \Phi \tilde{\mathcal{K}}(q, s), \quad (17)$$

with $\tilde{\mathcal{K}}(q, s) = \mathcal{K}(k, k_1, u)|_{k_1=qk, u=sk}$ (homogeneous of degree zero). All physical scales are absorbed in the prefactor $C_k^2 \epsilon k^{-5}$.

B. Gogoberidze (2007) Appendix A form

Using the Gaussian-erfc identity (Eq. A.1 of [2], $\int_0^\infty dy e^{-Ay^2 - B(y-x)^2} = \sqrt{\pi/[4(A+B)]} e^{-ABx^2/(A+B)} \text{erfc}[-Bx/\sqrt{A+B}]$), the ω_1 integration in Eq. (16) can be done in closed form, yielding the doubly-bounded form

$$H_{ijij}(k, \omega) = \frac{\varepsilon}{24\pi(2\pi)^{3/2}k} \int_{k_0}^{k_d} dk_1 k_1^{-10/3} \int_{\max(|k_1-k|, k_0)}^{\min(k_1+k, k_d)} du u^{-10/3} (k_1^{-4/3} + u^{-4/3})^{-1/2} \times \left[27 - \frac{k^2}{k_1^2} - \frac{k^2}{u^2} + \frac{k^4}{2k_1^2 u^2} + \frac{k_1^2}{2u^2} + \frac{u^2}{2k_1^2} \right] \exp\left(-\frac{2\varepsilon^{-2/3}\omega^2}{k_1^{4/3} + u^{4/3}}\right) \text{erfc}\left(-\frac{\sqrt{2}\varepsilon^{-1/3}\omega}{(k_1^{-4/3} + u^{-4/3})^{1/2}}\right). \quad (18)$$

k_0 is the outer scale and k_d the dissipation scale; the integration window collapses near the boundaries of the inertial range. This integral is finite as $k \rightarrow 0$.

Define the dimensionless parameters

$$M = \left(\frac{\varepsilon}{k_0}\right)^{1/3}, \quad \frac{k_d}{k_0} = R^{3/4}, \quad p = \frac{k}{k_0}, \quad q = \frac{\omega}{k_0}, \quad x = \left(\frac{k_1}{k_0}\right)^{-4/3}, \quad y = \left(\frac{u}{k_0}\right)^{-4/3}, \quad (19)$$

where $M = (\varepsilon/k_0)^{1/3}$ is the outer-scale turbulent Mach number (sound speed set to unity), equal to the rms eddy velocity u_0 . The Mach number enters the GW kernel through the k -dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}} \varepsilon^{1/3} k^{2/3} = \frac{M}{\sqrt{2\pi}} k_0^{1/3} k^{2/3}$, which couples frequency to wavenumber in the Gaussian factor of Eq. (18). As a result M sets the *effective* wavenumber support of the integral: the exponential $\exp[-2q^2/(M^2(z^{4/3} + y^{4/3}))]$ (with $z = k_1/k_0$) cuts off contributions with $z^{4/3} + y^{4/3} \lesssim 2q^2/M^2$, so the peak of the GW spectrum shifts with M . The substitution $k_1 = k_0 x^{-3/4}$, $u = k_0 y^{-3/4}$ gives $dk_1 k_1^{-10/3} = -(3/4)k_0^{-7/3} x^{3/4} dx$ and $(k_1^{-4/3} + u^{-4/3})^{-1/2} = k_0^{2/3} (x + y)^{-1/2}$; the integration limits become $x \in [R^{-1}, 1]$ and $y \in [y_{\min}(x), y_{\max}(x)]$ with

$$y_{\max}(x) = \min(|x^{-3/4} - p|^{-4/3}, 1), \quad y_{\min}(x) = \max((x^{-3/4} + p)^{-4/3}, R^{-1}). \quad (20)$$

Plugging into Eq. (18) yields

$$H_{ijij}(p, q) = \frac{3M^3 k_0^{-4}}{256(2\pi)^{3/2} p} \int_{R^{-1}}^1 dx x^{3/4} \int_{y_{\min}}^{y_{\max}} dy y^{3/4} (x + y)^{-1/2} \times \left[27 - p^2(x^{3/2} + y^{3/2}) + \frac{p^4}{2} x^{3/2} y^{3/2} + \frac{1}{2}(x^{-3/2} y^{3/2} + y^{-3/2} x^{3/2}) \right] \times \exp\left(-\frac{2xy}{x+y} \frac{q^2}{M^2}\right) \text{erfc}\left(-\frac{\sqrt{2}q}{M\sqrt{x+y}}\right). \quad (21)$$

C. Mach-dependent peak of the stationary spectrum

Evaluating the kernel (21) on the GW dispersion $\omega = k$ (the diagonal $q = p$ of Eq. (8)) gives the dimensionless energy-density spectrum $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p; M, R)$ of stationary Kraichnan-K41 turbulence. The Mach number

enters *only* through the sweeping rate $\eta_k \propto M k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ and the accompanying erfc in Eq. (21). This factor is a soft cutoff that suppresses the integrand once $p \gtrsim M$, so the spectral peak tracks the Mach number, $p_{\text{peak}} \sim M$, until it saturates near the outer scale at large M . Figure 1 shows this directly: the peak slides from $p_{\text{peak}} \simeq 0.05$ at $M = 0.03$ to $p_{\text{peak}} \simeq 2.6$ at $M = 3$, while the amplitude climbs steeply (the M^3 prefactor of Eq. (21) times the widening M -support of the integral). The infrared rise is the universal causal slope $\Omega_{\text{GW}} \propto p^3$ (Sec. IC), common to all M ; only the location and height of the peak respond to M .

The M -dependence of the IR amplitude is fixed entirely by the prefactor: for $p \ll M$ the Gaussian and erfc factors of Eq. (21) reduce to unity, so $H_{ijij}(p \rightarrow 0)$ is set by the M^3 prefactor alone and $\Omega_{\text{GW}} \rightarrow C M^3 p^3$ with C a universal (M -independent) constant. Equivalently the characteristic strain scales as $h_c = \sqrt{p} \bar{H} \sim M^{3/2} \sqrt{p}$ in the IR. Compensating the spectrum by M^{-3} (the strain by $M^{-3/2}$) therefore collapses every curve onto the single universal p^3 line in the IR, isolating the genuinely M -dependent peak and UV; Fig. 1(b) confirms this collapse numerically (residual spread $< 1.5\%$ at $p \simeq 3 \times 10^{-4}$, and a measured IR exponent $d \ln H / d \ln M \rightarrow 3.00$ as $p \rightarrow 0$).

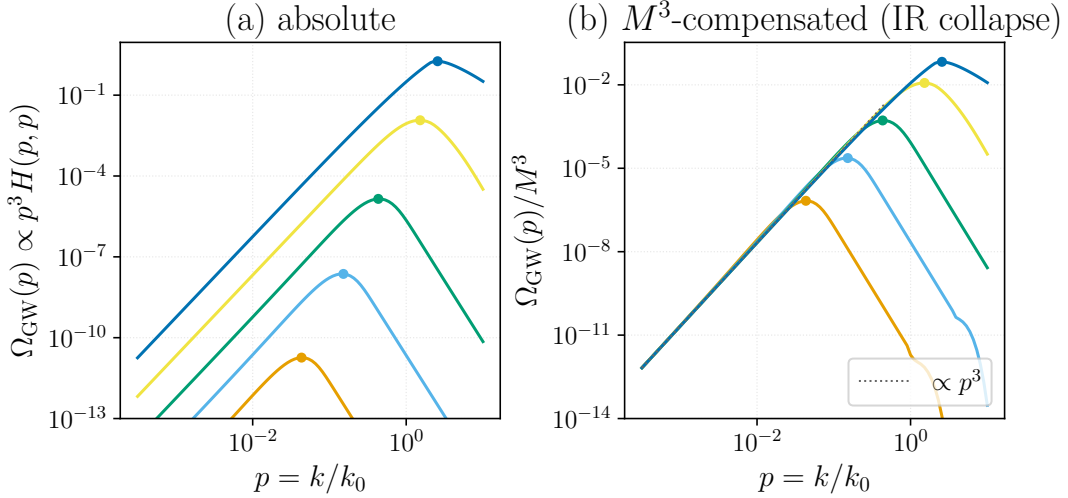


FIG. 1. Stationary Kraichnan-K41 (Gogoberidze 2007) GW spectrum $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p; M, R)$, computed from Eq. (21) on the on-shell diagonal of Eq. (8), at $R = 10^4$. The five curves are $M = 0.03, 0.1, 0.3, 1, 3$, ordered by increasing amplitude and peak frequency (lowest, leftmost curve is $M = 0.03$; highest, rightmost is $M = 3$); dots mark each peak. (a) Absolute: shared IR slope p^3 , amplitude growing $\sim M^3$, peak moving right with M ($p_{\text{peak}} \sim M$ in the inertial range, bending over near the outer scale). (b) M^3 -compensated, Ω_{GW}/M^3 : since the IR amplitude scales as M^3 (strain as $M^{3/2}$), all curves collapse onto the universal p^3 line (dotted) in the IR and peel off only at their M -dependent peak. Generated by `Notebooks/stationary_mach_peak.py`.

D. Spectra in the $\Omega(k)/k$ convention: comparison with simulations

To compare the analytic kernel with the direct numerical simulations of Roper Pol et al. [6] we recast the spectra in their convention. There $\Omega(k)$ denotes the energy *per logarithmic wavenumber*, $\Omega(k) = d\rho/d \ln k$, so that $\Omega(k)/k = E(k)$ is the ordinary per- dk spectrum. Our master result $\Omega_{\text{GW}}(p) \propto p^3 H_{ijij}(p, p)$ of Eq. (8) is precisely $d\rho_{\text{GW}}/d \ln k$, hence

$$\frac{\Omega_{\text{GW}}(k)}{k} \propto \frac{\Omega_{\text{GW}}(p)}{p} = p^2 H_{ijij}(p, p; M, R), \quad p = \frac{k}{k_0}, \quad (22)$$

with H_{ijij} the stationary kernel of Eq. (21). To compare on the simulations' own footing we feed a *full* fluid input spectrum into this kernel, with both an infrared band and the Kolmogorov inertial range,

$$E(k) \propto \begin{cases} k^4, & k < k_0 \quad (\text{Batchelor infrared band}), \\ k^{-5/3}, & k \geq k_0 \quad (\text{Kolmogorov inertial range}), \end{cases} \quad (23)$$

joined continuously at k_0 . The spectrum enters the Appendix-A integral Eq. (18) only through the amplitude $A(k) = E(k)/4\pi k^2$; writing $A(k) = S(k) A_{\text{Kol}}(k)$ with the Kolmogorov law $A_{\text{Kol}} \propto k^{-11/3}$ extrapolated to all k , the shape factor is $S(k) = 1$ for $k \geq k_0$ and $S(k) = (k/k_0)^{s+5/3}$ for $k < k_0$ (energy slope $E \propto k^s$, here $s = 4$). Every kinematic

and temporal factor of the kernel is left untouched: we extend the k_1, u integrations down to an infrared cutoff $k_{\text{IR}} = k_0/R_{\text{IR}}$ and multiply the integrand by $S(k_1)S(u)$. Setting $R_{\text{IR}} = 1$ recovers Eq. (21) identically.

Figure 2 shows the resulting GW spectrum at $M = 1$ with $R_{\text{IR}} = 100$ (two infrared decades), together with the full fluid input spectrum and the two slopes the literature reports for the GW inertial range, drawn as reference guides: $k^{-11/3}$ from the simulation of [6] (run ini2) and $k^{-9/2}$ from [2]. We plot no simulation data points and defer a quantitative data-level comparison to future work. Both curves are *dimensionless*: the GW curve is the bare kernel with the source stress normalised to unity, so the GW–fluid vertical offset carries no physical meaning (the relic Ω_{GW} acquires the additional suppression $\sim \Omega_{\text{M}}^2(\mathcal{H}_*/k_*)$ and gravitational couplings that the kernel omits, which is why the simulated $\Omega_{\text{GW}}/k \sim 10^{-13}$ lies far below the fluid spectrum); only slopes and peak locations are comparable, and the fluid curve is rescaled to share the GW peak.

Three features survive this honest treatment. (i) *The infrared stays causal.* The GW spectrum rises with the causal slope $\Omega_{\text{GW}} \propto k^3$ ($\Omega_{\text{GW}}/k \propto k^2$, fitted $k^{+2.0}$); it does *not* inherit the steeper fluid k^4 band, because the GW infrared is fixed by the analyticity of the stress rather than by the input spectrum. (ii) *The infrared band lifts the amplitude, not the inertial range.* Feeding the band raises the GW amplitude near and below the peak by a factor $\simeq 1.9$ and shifts the peak from $k \simeq 1.5 k_0$ to $k \simeq k_0$, a change that depends only weakly on the infrared slope (a further $\simeq 1.4$ between Batchelor k^4 and Saffman k^2) and leaves the inertial range untouched. (iii) *The ultraviolet tension is robust.* Above the peak the spectrum steepens *continuously*, and at $M = 1$ more steeply than either reference: the measured post-peak slope is $\simeq k^{-5.0}$ —steeper than both Gogoberidze’s $k^{-9/2}$ and the simulation’s $k^{-11/3}$ —before the k -dependent sweeping decorrelation bends it below any single power law. The analytic stationary kernel therefore does *not* reproduce the extended $k^{-11/3}$ range seen in the simulation; at the simulations’ Mach number it is markedly steeper, crossing that slope rather than following it, and this conclusion is unchanged by the infrared treatment. This high- k disagreement is precisely the simulation–analytic tension that motivates this paper—and that the decaying kernel of Sec. III substantially relieves.

E. Discussion: the Mach number, equivalent definitions, and the spectral shape

The preceding two subsections make the Mach number the single control parameter of the stationary spectrum, so it is worth stating precisely what M is, why the conflicting definitions in the literature are the *same* quantity, and what about the spectrum does and does not respond to it.

a. *Explicit formulation.* Throughout we use

$$M = \left(\frac{\varepsilon}{k_0}\right)^{1/3} = u_0, \quad (c_s \equiv 1), \quad (24)$$

the outer-scale eddy velocity in units of the sound speed (Eq. (19)). The grouping is fixed by dimensions: with $[\varepsilon] = L^2 T^{-3}$ and $[k_0] = L^{-1}$ one has $\varepsilon/k_0 = (L/T)^3$, so $(\varepsilon/k_0)^{1/3}$ is a velocity and the inverse grouping $(k_0/\varepsilon)^{1/3}$ — which appears occasionally in the literature — is an *inverse* velocity and cannot be a Mach number. M does not enter Eq. (18) as a free amplitude; it enters *only* through the k -dependent sweeping rate $\eta_k = \frac{1}{\sqrt{2\pi}} \varepsilon^{1/3} k^{2/3} = \frac{M}{\sqrt{2\pi}} k_0^{1/3} k^{2/3}$, hence only through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ and the accompanying erfc of Eq. (21).

b. *Two definitions, one quantity.* The literature quotes the Mach number either as a velocity ratio $M = u_{\text{rms}}/c_s$ or as the cascade combination $(\varepsilon/k_0)^{1/3}$ [2]. For fully developed Kolmogorov turbulence these are the same quantity: setting $c_s = 1$ and using the inertial-range relation $\varepsilon \sim u_0^3 k_0$ turns the first into the second. The equivalence rests on three assumptions: (i) a stationary, constant-flux cascade with a well-defined ε across an inertial range of width $k_d/k_0 = R^{3/4} \gg 1$; (ii) energy concentrated at the outer scale, which the $E(k) \propto k^{-5/3}$ slope guarantees because $\int_{k_0}^{\infty} E dk = \frac{3}{2} C_K (\varepsilon/k_0)^{2/3}$ converges at k_0 ; and (iii) absorbing the resulting $\mathcal{O}(1)$ constant, so that $u_{\text{rms}} = \sqrt{\frac{3}{2} C_K} (\varepsilon/k_0)^{1/3} \approx (1.5\text{--}2) (\varepsilon/k_0)^{1/3}$ depending on C_K and on whether one normalises $\int E dk$ to $\langle u^2 \rangle$ or $\frac{1}{2} \langle u^2 \rangle$. The bare $(\varepsilon/k_0)^{1/3}$ is thus the single-eddy outer velocity, smaller than the integrated rms by this convention-dependent factor — the dominant source of numerical disagreement between papers for the same physical flow. The two definitions *diverge* when the assumptions fail: for non-Kolmogorov spectra in which the energy integral is ultraviolet-dominated (the white-noise / Batchelor model of Sec. IV B, $E \propto k^2$), where $(\varepsilon/k_0)^{1/3}$ no longer measures u_{rms} ; for decaying turbulence, where $\varepsilon \rightarrow \varepsilon(t)$ and $M \rightarrow M(t)$ (Sec. V); and in MHD, where the relevant ratio is often the Alfvénic $M_A = u_{\text{rms}}/v_A$ rather than the acoustic M [3, 7].

c. *Shape versus peak versus amplitude.* It is essential to separate three properties of the spectrum, only one of which is M -independent. (i) *Shape.* The dimensionless spectral form — the causal infrared slope $\Omega_{\text{GW}} \propto p^3$ (Sec. IC) and the inertial power law — is universal: it is set by causality and by the source power law, not by M . This is the property that is legitimately called “ M -independent,” and it is what the M^{-3} -compensated collapse of Fig. 1(b)

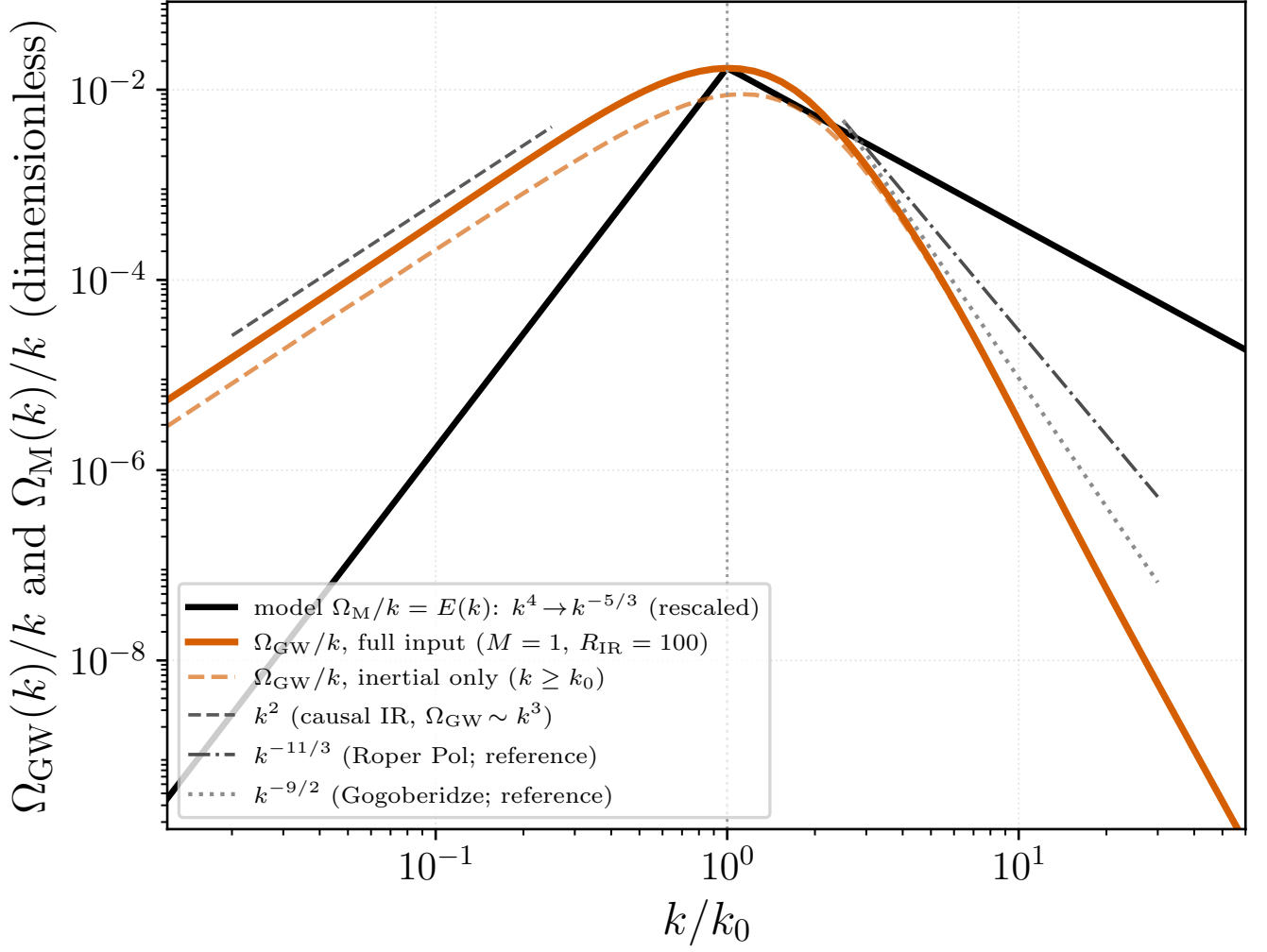


FIG. 2. Analytic spectra in the Roper Pol et al. [6] convention $\Omega(k)/k = E(k)$. Orange (solid): GW spectrum $\Omega_{\text{GW}}(k)/k = p^2 H_{ijij}(p, p; M, R)$ [Eq. (22)] computed from a *full* fluid input spectrum—an infrared band $E \propto k^4$ ($k < k_0$) joined to the Kolmogorov range $E \propto k^{-5/3}$ ($k \geq k_0$), Eq. (23)—fed into the Appendix-A kernel Eq. (18) through the shape factor $S(k) = A/A_{\text{Kol}}$ ($M = 1$, $R = 10^4$, infrared range $R_{\text{IR}} = k_0/k_{\text{IR}} = 100$). Orange (dashed): the same kernel restricted to the inertial range only ($R_{\text{IR}} = 1$, i.e. core Eq. (21))—the infrared band raises the amplitude near the peak but leaves the ultraviolet unchanged. Black: the full fluid input spectrum, rescaled to share the GW peak (the GW–fluid vertical offset is unphysical; only shapes compare). Grey dashed: the causal infrared guide k^2 ($\Omega_{\text{GW}} \sim k^3$). Dash-dot/dotted: the $k^{-11/3}$ (Roper Pol, run ini2) and $k^{-9/2}$ (Gogoberidze; reference) slopes *reported* in the literature, as references only—no simulation data points are plotted. At $M = 1$ the GW spectrum steepens past the peak to $\simeq k^{-5.0}$, steeper than both references, independently of the infrared treatment. Generated by `Notebooks/roperpol_convention_spectra.py`.

isolates. (ii) *Peak location*. Here the literature genuinely splits. In spatial-scale-anchored treatments the peak is pinned to the source scale k_0 (set by the largest eddy / bubble size), so $p_{\text{peak}} = \mathcal{O}(1)$ and M controls only the amplitude [4]; in sweeping-anchored treatments the peak is tied to the eddy-turnover frequency $u_0 k_0 \propto M k_0$, so it shifts with M [2, 6]. Our kernel realises the second picture through $\eta_k \propto M k^{2/3}$, giving $p_{\text{peak}} \sim M$ (Sec. II C). The two are not in contradiction: a peak “at the stirring scale” is M -independent only because it is measured in units of k_0 and the source is assumed quasi-static over an eddy. (iii) *Total energy / amplitude*. This is never M -independent. In the stationary kernel the infrared amplitude scales as the M^3 prefactor of Eq. (21) ($\Omega_{\text{GW}} \rightarrow C M^3 p^3$, Sec. II C); more generally the quadrupole argument gives $\Omega_{\text{GW}} \propto \Omega_K^2 \propto M^4$, steepening toward M^{5-6} once the finite source duration is included [4], the extra power being absent from a single-time stationary kernel. We derive this progression explicitly for the decaying source in Sec. III A: the full-spatial kernel gives an infrared amplitude $\propto M^4$ and a source-pinned peak amplitude $\propto M^{5.3}$ [Eq. (30), Fig. 6]. Consequently a stated “ M -independence” of the GW signal can only refer

to the shape (i); any claim that the peak position and the total energy are simultaneously M -independent is either a normalised-units artifact (dividing by Ω_K^2 and measuring p in units of k_0) or a conflation of the universal shape with the dimensional peak and amplitude.

III. LOCATION OF THE SPECTRAL PEAK: SWEEPING VERSUS SOURCE SCALE

A striking feature of the stationary spectrum of Fig. 1 is that, for subsonic turbulence, the GW energy density peaks at wavenumbers *far below* the outer scale, $p_{\text{peak}} = k_{\text{peak}}/k_0 \ll 1$ (e.g. $p_{\text{peak}} \simeq 0.04$ at $M = 0.03$). This is counterintuitive: the GW source is the quadratic stress $T_{ij} \sim u_i u_j$, and for a velocity field concentrated at the stirring scale k_0 one expects the stress—and hence the radiation—to peak near $k \sim k_0$, with spatial support extending to at most $k = 2k_0$ by the triangle inequality. We show here that the analytic peak is fixed instead by the *temporal* (sweeping) decorrelation, giving $p_{\text{peak}} \simeq 1.47 M$, and that the “ $k \simeq 2k_0$ ” expectation is recovered only in the transonic limit.

a. A one-parameter scaling. In the dimensionless kernel of Eq. (21) the Mach number enters *only* through the sweeping rate $\eta_k \propto M k^{2/3}$, i.e. through the Gaussian $\exp[-\frac{2xy}{x+y} q^2/M^2]$ (and its accompanying erfc) evaluated on the GW cone $q = p$ of Eq. (8). The geometric kernel and the triangle-inequality integration bounds carry no M . The on-cone spectrum must therefore take the self-similar form

$$\Omega_{\text{GW}}(p; M) = A M^3 p^3 G\left(\frac{p}{M}\right), \quad (25)$$

in which $A M^3 p^3$ is the universal causal infrared rise (Sec. IC, with the M^3 prefactor of Eq. (21)) and G is a cutoff of the single combination $\xi \equiv p/M$, with $G(0) = 1$ and $G(\xi \rightarrow \infty) \rightarrow 0$.

b. Numerical proof of the collapse. Equation (25) is verified directly in Fig. 3. Panel (b) plots $\Omega_{\text{GW}}/(M^3 p^3)$ against $\xi = p/M$ for $M = 0.03$ –3: the curves collapse onto a single cutoff $A G(\xi)$, with infrared plateau $A \simeq 2.1 \times 10^{-2}$ and $< 13\%$ residual spread dominated by the transonic $M = 3$ case. The decorrelation cutoff thus depends on p and M *only* through p/M —the unambiguous signature of the sweeping rate—and switches on at $\xi \sim 1$, i.e. $p \sim M$.

c. Peak condition and raw peak location. Differentiating Eq. (25) on the cone,

$$\frac{d \ln \Omega_{\text{GW}}}{d \ln p} = 3 + \frac{d \ln G}{d \ln \xi} = 0 \implies \left. \frac{d \ln G}{d \ln \xi} \right|_{\xi_*} = -3, \quad p_{\text{peak}} = \xi_* M, \quad (26)$$

so the peak sits where the cutoff has logarithmic slope -3 , at a fixed ξ_* independent of M . Reading the peak directly from each computed spectrum (Fig. 3c; sub-grid argmax, no model fit) gives a location linear in M across the entire subsonic–transonic range,

$$p_{\text{peak}} = 1.48 M^{1.00} \quad (M \lesssim 1), \quad (27)$$

with $\xi_* = 1.47$ read from the collapsed shape, consistent with Eq. (26). Above $M \simeq 1$ the peak runs into the outer scale and the linear law saturates: p_{peak}/M falls from 1.49 at $M = 1$ to 0.77 at $M = 3.4$. The peak reaches the “source-scale” value $p \simeq 2$ only at transonic $M \simeq 1.4$.

d. Resolution. The naive “ $k \simeq 2k_0$ ” is a purely *spatial* statement: it is the largest wavenumber at which the quadratic stress has support, the triangle edge $\Theta(2-p)$ that appears explicitly in the monochromatic kernel [Eq. (35)]. It is an ultraviolet *edge*, not a peak. The GW peak, by contrast, is fixed in *frequency*: gravitational waves are sourced on the cone $\omega = k$, while a turbulent eddy of wavenumber k decorrelates at the sweeping frequency $\eta_k \sim M k_0^{1/3} k^{2/3}$, which for subsonic flow is $\ll k$ for every $k \gtrsim k_0$. The source therefore has negligible temporal power at the frequency $\omega = k$ needed to radiate at wavenumber k , except at the lowest wavenumbers $k = \omega \lesssim M k_0$, and the spectrum is throttled down to $p_{\text{peak}} \sim M$. This is the aeroacoustic (Lighthill) suppression of slow sources: subsonic turbulence radiates at its eddy-turnover frequency $\sim M k_0$, not at its spatial scale k_0 . Whether the true peak follows this sweeping scale or remains pinned near k_0 (Sec. IIE) is precisely the simulation–analytic tension this paper addresses; the stationary kernel commits unambiguously to the sweeping scale.

e. Caveat: stationarity versus the simulated peak. The scaling $p_{\text{peak}} \simeq 1.47 M$ derived above is specific to *stationary*, continuously stirred turbulence, in which the source at scale k carries temporal power only up to its eddy frequency η_k . Cosmological turbulence is not of this kind: it is sourced *impulsively*—e.g. at a phase transition—and then decays. A sudden onset of duration Δt injects broadband temporal power up to $\omega \sim 1/\Delta t \sim k_0$, which the single-time stationary kernel does not contain. Because the spatial stress is far more powerful at the outer scale k_0 than at $M k_0$, this broadband component pins the GW peak near the *spatial* stress scale, $p \sim 1$ –2, almost independently of M . This is what the direct simulations of [6] display, and it is the spatial-scale picture of Sec. IIE. The two predictions

therefore *coincide* in the transonic band $M \sim 1$, where all current simulations sit, and *diverge* only in the subsonic corner $M \ll 1$, which no simulation has yet probed (Fig. 4). This peak displacement is the spatial counterpart of the amplitude deficit noted in Sec. II E—the stationary kernel gives $\Omega_{\text{GW}} \propto M^4$ rather than the M^{5-6} recovered once the finite source duration is restored—both stemming from the broadband temporal power that an impulsive, finite-duration source has and the stationary kernel omits. We compute the decaying case directly in Sec. III A below; it confirms a source-scale peak $p \simeq 2.4$, nearly independent of M , but traces it to the *shape* of the decaying temporal correlation rather than to the impulsive onset emphasised here.

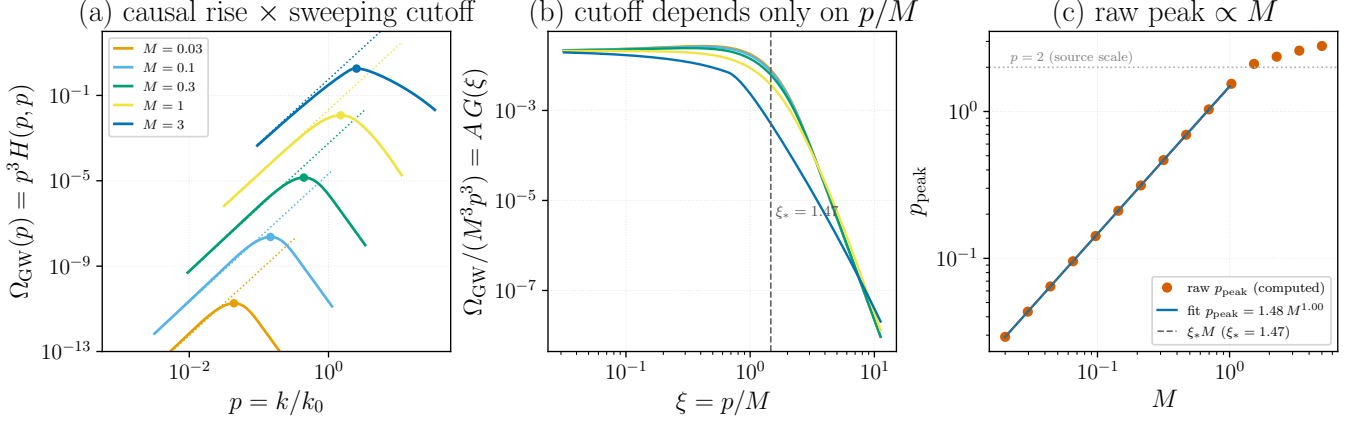


FIG. 3. Why the stationary GW spectrum peaks at $p \simeq 1.47 M$ rather than at the source scale $p \sim 2$, computed from $\Omega_{\text{GW}}(p) = p^3 H_{ijij}(p, p; M, R)$ [Eq. (21), $R = 10^4$]. (a) Each spectrum (solid) follows its causal infrared asymptote $A M^3 p^3$ (dotted) and rolls over at the peak (dot) where the sweeping cutoff switches on. (b) Collapse: $\Omega_{\text{GW}}/(M^3 p^3)$ versus $\xi = p/M$ falls on a single curve $AG(\xi)$ for every M , proving the cutoff depends only on p/M [Eq. (25)]; the universal peak position is $\xi_* = 1.47$. (c) Peak location read directly from each computed spectrum (points) versus M : a power law $p_{\text{peak}} = 1.48 M^{1.00}$ (line) [Eq. (27)], coincident with $\xi_* M$ (dashed), that bends below it toward the $p = 2$ source scale (dotted) only at transonic M . Generated by `Notebooks/stationary_peak_analysis.py`.

A. The decaying source pins the peak at the source scale

The caveat above is borne out quantitatively by repeating the peak analysis with the *decaying* temporal model in place of the stationary one. We keep the same full-spatial kernel—identical geometric factor and triangle-inequality integration bounds—but replace the Kraichnan Gaussian decorrelation on each stress leg by the BK2016 power law $(1 + |\tau|/\tau_k)^{-2/3}$ of Sec. IV A 2, with the sweeping eddy time $\tau_k = \sqrt{2\pi}/(M k_0^{1/3} k^{2/3})$.

a. Method: Fourier transform of the time-domain product. The temporal factor of the decaying kernel is the frequency convolution $\int d\omega_1 \hat{g}(\omega_1 \tau_{k_1}) \hat{g}((\omega - \omega_1) \tau_u)$ of the two legs. By the convolution theorem this equals the Fourier transform of the time-domain *product* of the two decorrelations,

$$\int d\omega_1 \hat{g}(\omega_1 \tau_1) \hat{g}((\omega - \omega_1) \tau_2) = \frac{2\pi}{\tau_1 \tau_2} \int_0^\infty d\tau \cos(\omega \tau) \left(1 + \frac{\tau}{\tau_1}\right)^{-2/3} \left(1 + \frac{\tau}{\tau_2}\right)^{-2/3}, \quad (28)$$

with $\tau_1 = \sqrt{x}/M$, $\tau_2 = \sqrt{y}/M$. The right-hand side, unlike the convolution—whose integrand is integrably singular at both endpoints—is smooth and rapidly convergent; evaluated this way the kernel is some two orders of magnitude cheaper to compute and well converged.

b. Result: an M -independent, source-scale peak. On the GW cone the decaying spectrum $\Omega_{\text{GW}}(p) = p^3 H_{ijij}^{\text{decay}}(p, p; M, R)$ peaks at

$$p_{\text{peak}}^{\text{decay}} \simeq 2.4, \quad (29)$$

essentially independent of both the Mach and the Reynolds number: the peak moves only from 2.29 to 2.56 as M runs over 0.3–2, and from 2.40 to 2.60 as R runs over 10^3 – 10^5 (a peak set by the dissipation scale would scale as $R^{3/4}$). This is in sharp contrast to the stationary law $p_{\text{peak}} = 1.47 M$ of Eq. (27), which tracks the sweeping scale. The two are compared in Fig. 5.

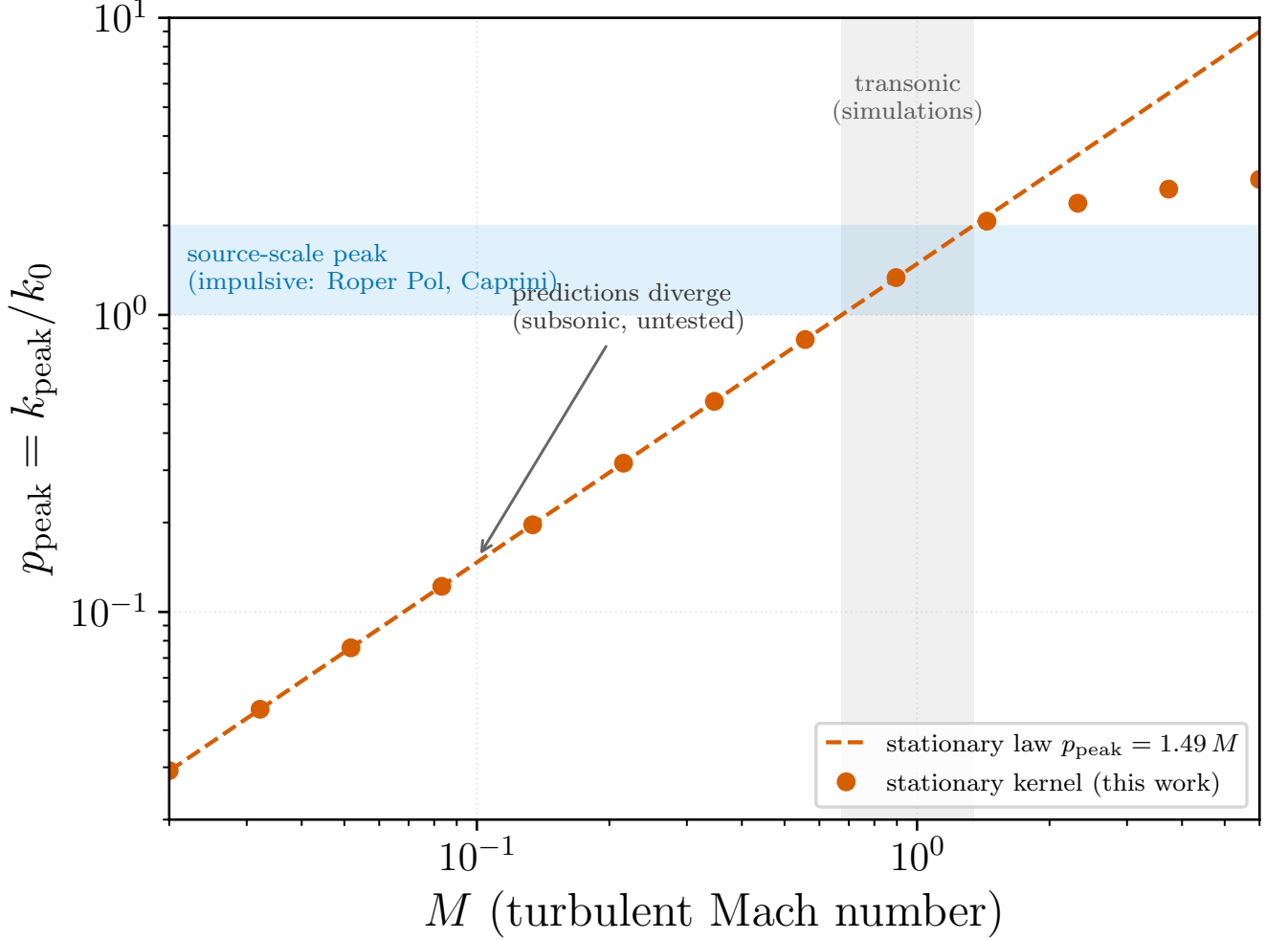


FIG. 4. Reconciliation of the stationary and impulsive pictures of the GW peak. Points and dashed line: the peak $p_{\text{peak}} = k_{\text{peak}}/k_0$ computed from the stationary kernel $\Omega_{\text{GW}} = p^3 H_{ijij}(p, p; M, R)$ ($R = 10^4$), following $p_{\text{peak}} \simeq 1.5 M$. Shaded horizontal band ($p \sim 1$ – 2): the competing spatial-scale picture, in which an impulsively sourced, decaying flow radiates at the stress scale independently of M (Roper Pol et al. [6]; Caprini et al. [4])—a schematic, *not* computed here and not data. The two coincide in the transonic band (vertical strip), where current simulations sit, and diverge only for subsonic $M \ll 1$. Generated by `Notebooks/peak_reconciliation.py`.

c. Mechanism: the high-frequency tail. The peak is fixed by the high-frequency tail of the temporal correlation. The stationary Gaussian sweeping factor $\exp[-\frac{2xy}{x+y}q^2/M^2]$ cuts the source off sharply once $q \gtrsim M$, throttling the spectrum down to $p \sim M$. The power-law decorrelation of decaying turbulence instead has a *heavy* frequency tail, $\hat{g}(q) \sim q^{-1/3}$, so the source retains power at the frequency $\omega = k$ required to radiate at every wavenumber k ; the spectrum then extends past the sweeping cutoff and its peak is fixed by the spatial structure of the stress near the source scale $p \simeq 2.4$, independently of M . This realises the spatial-scale picture of Sec. II E and matches the source-scale peak of the decaying simulations [6]; the apparent contradiction with the low Gogoberidze peak dissolves once one recognises that the two correspond to opposite limits of the temporal correlation—Gaussian (rapidly decorrelating, stationary) versus power law (slowly decorrelating, decaying).

d. The decorrelation shape, not the onset. This pinning is already present in the *quasi-stationary* decaying kernel used here, in which the two-time correlation is taken at a single emission epoch and there is no impulsive switch-on. It is therefore the *shape* of the temporal decorrelation—its heavy frequency tail—rather than the broadband power of a sudden onset invoked in the caveat of Sec. III, that is responsible for displacing the peak to the source scale. Restoring the finite source duration (the self-similar slow-time integration) modifies the amplitude and overall normalisation but leaves this peak displacement intact.

e. Amplitude: an extra power of M . The same kernel fixes the amplitude as cleanly as the peak. The Mach number enters the full-spatial decaying kernel in only two places: the prefactor $\propto M^3$ of Eq. (21) and the temporal factor Eq. (28). Rescaling $\tau \rightarrow \tau/M$ in the latter shows it equals $M F(\omega/M)$ with F independent of M (the geometric weight, kernel bracket and triangle bounds carry no M), so on the GW cone $\omega = k$ the spectrum factorises exactly as

$$\Omega_{\text{GW}}(p; M) = M^4 p^3 \Psi(p, p/M), \quad (30)$$

with Ψ the M -independent rescaled spatial integral. Two limits follow. In the infrared $p/M \rightarrow 0$ freezes $\Psi \rightarrow \Psi(p, 0)$, and the amplitude scales as the bare prefactor, $\Omega_{\text{GW}} \propto M^4$ (measured: $M^{3.98}$)—one power steeper than the stationary M^3 of Eq. (21), the extra power being the single M from the temporal convolution. At the source-pinned peak $p \simeq 2.4$, by contrast, the argument p/M slides down the universal Ψ as M grows, and the scaling steepens to $\Omega_{\text{GW}}^{\text{peak}} \propto M^{5.3}$, with the band-integrated energy following the same $M^{5.3}$ (Fig. 6). This *derives*, rather than postulates, the $M^4 \rightarrow M^{5.6}$ progression quoted in Sec. II E: the M^4 is the genuine infrared prefactor, while the $M^{5.6}$ peak and energy steepening follows from the source-scale pinning acting on the p/M -dependent temporal weight—a mechanism already present in the quasi-stationary kernel, distinct from and prior to the finite-duration correction. (For reference the stationary peak amplitude scales similarly, $\sim M^{5.4}$, because its peak rides outward to $p = 1.47M$; the amplitude discriminant between the two pictures is therefore the *infrared* slope, M^4 versus M^3 , not the peak height.)

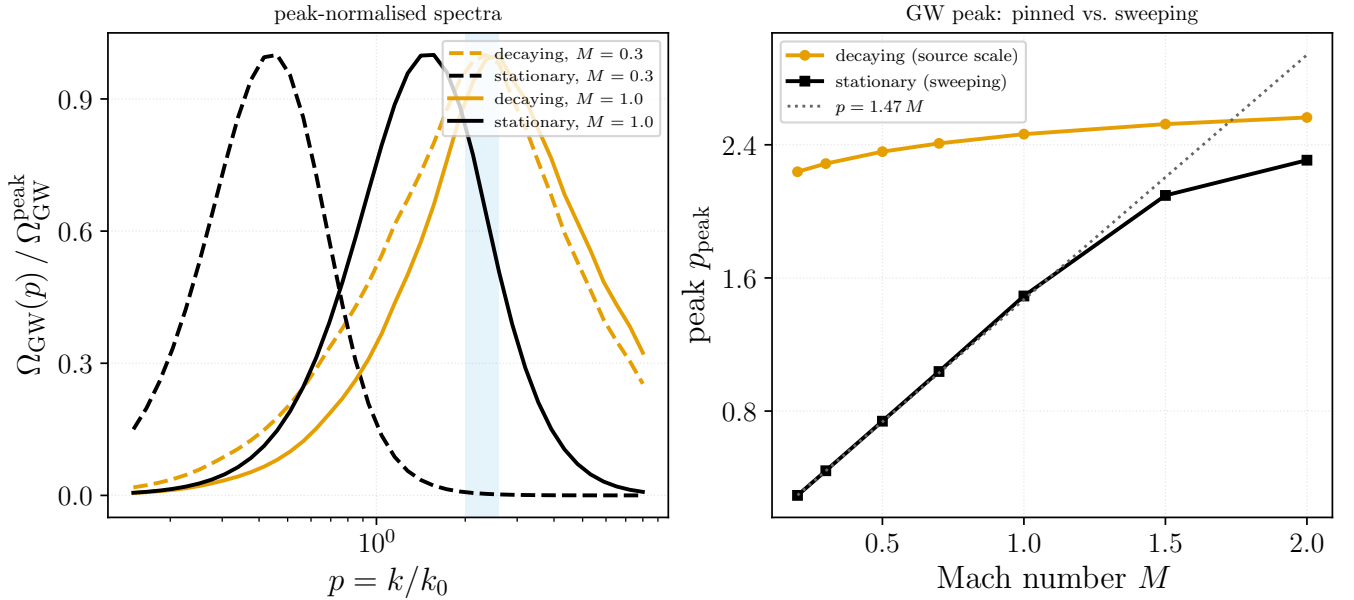


FIG. 5. The decaying source pins the GW peak at the source scale, in contrast to the stationary sweeping peak. (a) Peak-normalised spectra $\Omega_{\text{GW}}(p) = p^3 H_{ijij}(p, p; M, R)$ ($R = 10^4$) for the decaying (power-law, orange) and stationary (Gaussian, black) kernels at $M = 0.3$ (dashed) and $M = 1$ (solid): the decaying peaks fall in the source-scale band $p \simeq 2$ – 2.6 (shaded) for both M , whereas the stationary peaks shift with M . (b) Peak location versus M : the decaying peak is flat at $p_{\text{peak}} \simeq 2.4$ [Eq. (29)], while the stationary peak follows $p_{\text{peak}} = 1.47 M$ (dotted) until it saturates near transonic M . Computed with the full-spatial kernel via the FT-of-product Eq. (28); generated by `Notebooks/fullspatial_decay.py`.

IV. MONOCHROMATIC AND WHITE-NOISE SPECTRA

We now consider two alternative models for the turbulence power spectrum and carry out the same GW-kernel derivation in each case. The first model replaces the Kolmogorov spectrum by a monochromatic (delta-function) distribution in wavenumber; the second takes the real-space two-point function to be spatially uncorrelated (delta function in $\mathbf{r}$), which corresponds to a flat (white-noise) spectrum in k -space. For each spatial model we work out both the Kraichnan stationary decorrelation and the decaying-turbulence temporal model, giving four cases in total.

The derivation follows the same sequence of steps as in Sec. II: (i) write H_{ijij} as a double convolution in both $\mathbf{k}$ and ω , (ii) insert the spectral model, (iii) perform the tensor contractions to get $\mathcal{K}$, (iv) switch to spherical coordinates and the variable $u = |\mathbf{k} - \mathbf{k}_1|$, (v) evaluate or simplify the temporal integral, and (vi) introduce dimensionless variables.

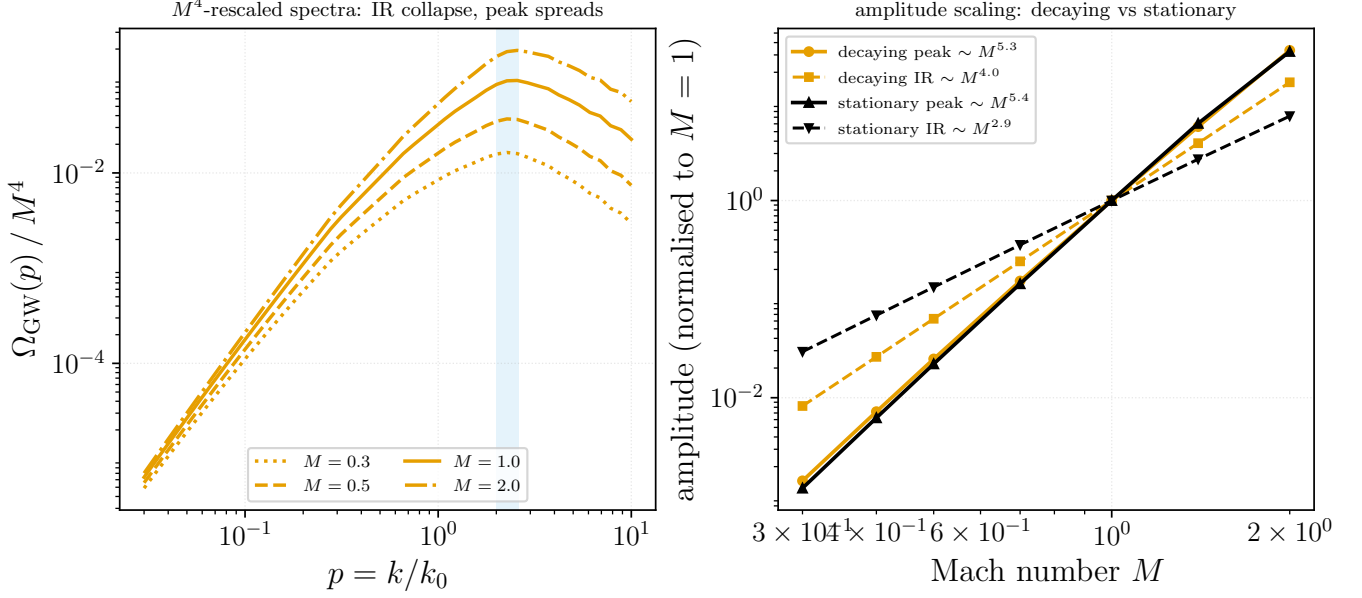


FIG. 6. Mach scaling of the full-spatial decaying GW amplitude, from the exact factorisation $\Omega_{\text{GW}} = M^4 p^3 \Psi(p, p/M)$ [Eq. (30)]. (a) M^4 -rescaled spectra $\Omega_{\text{GW}}(p)/M^4$ for $M = 0.3$ – 2 ($R = 10^4$): the curves collapse in the infrared—confirming the M^4 prefactor—and spread upward through the source-scale band (shaded), the peak amplitude growing faster than M^4 . (b) Amplitude versus M , normalised at $M = 1$, with fitted power laws: the decaying infrared amplitude scales as $M^{4.0}$ (one power above the stationary $M^{2.9}$), while the source-pinned peak amplitude steepens to $M^{5.3}$. Generated by `Notebooks/decaying_amplitude.py`.

A. Monochromatic energy spectrum $E(k) = E_0 \delta(k - k_0)$

Set $E(k) = E_0 \delta(k - k_0)$ so that $A(k) = E_0 \delta(k - k_0)/(4\pi k^2)$ and the spectral tensor reads

$$\tilde{F}_{ij}(\mathbf{k}, \omega) = \frac{E_0 \delta(k - k_0)}{4\pi k^2} P_{ij}(\hat{k}) \tilde{f}(\omega), \quad \int_0^\infty E(k) dk = E_0, \quad (31)$$

with $\tilde{f}(\omega)$ to be specified by the decorrelation model.

Starting from Eq. (13) in spherical coordinates, the radial integral collapses $k_1 \rightarrow k_0$ via $\delta(k_1 - k_0)$, fixing $u = \sqrt{k^2 + k_0^2 - 2kk_0\mu}$. Trading μ for u (azimuth contributes 2π) and using $\delta(u - k_0)$ from $A(u)$ yields the closed-form spatial reduction

$$2\pi \int_{-1}^1 d\mu A(u) = \frac{E_0}{2kk_0^2} \Theta(2k_0 - k), \quad (32)$$

with the Heaviside factor enforcing the triangle inequality $p \equiv k/k_0 \leq 2$. The geometric kernel at $k_1 = u = k_0$ evaluates to

$$\mathcal{K}_0(p) \equiv \mathcal{K}(k, k_0, k_0) = \frac{14}{3} - \frac{p^2}{3} + \frac{p^4}{12}. \quad (33)$$

Substituting Eqs. (32)–(33) in Eq. (13),

$$H_{ijij}(\mathbf{k}, \omega) = \frac{E_0^2 \Theta(2k_0 - k)}{2(2\pi)^8 k k_0^2} \mathcal{K}_0(p) I_\omega(\omega), \quad I_\omega(\omega) \equiv \int d\omega_1 \tilde{f}(\omega_1) \tilde{f}(\omega - \omega_1). \quad (34)$$

All spatial structure has been resolved in closed form; only the temporal convolution I_ω remains. Because both wavevectors are pinned to k_0 by the two δ -functions, there is no leftover wavenumber integration: the only spatial constraint is the M -independent triangle inequality $\Theta(2 - p)$. The Mach number therefore cannot reshape any spatial integration *bounds* here; it enters solely through the temporal decorrelation rate at the fixed scale k_0 , which we tie to M via $\eta_{k_0} = \frac{M}{\sqrt{2\pi}} k_0$ (cf. Eq. (19)). This contrasts with the white-noise case of Sec. IV B, where M does enter the surviving wavenumber integral.

1. Kraichnan decorrelation

For $\tilde{f}(\omega) = (2/\eta_0)e^{-\omega^2/\pi\eta_0^2}$ with $\eta_0 \equiv \eta_{k_0} = Mk_0/\sqrt{2\pi}$, the Gaussian convolution gives $I_\omega(\omega) = (2\sqrt{2\pi}/\eta_0)e^{-\omega^2/(2\pi\eta_0^2)}$. With $q \equiv \omega/k_0$ the exponent becomes $\omega^2/(2\pi\eta_0^2) = q^2/M^2$, so Eq. (34) separates into a dimensional prefactor and the boxed dimensionless kernel

$$H_{ijij}(\mathbf{k}, \omega) = \frac{\sqrt{2} E_0^2}{(2\pi)^7 k k_0^2 \eta_0} \mathcal{K}_0(p) \Theta(2-p) e^{-q^2/M^2}, \quad \boxed{\mathfrak{H}_{\text{Kraichnan}}^{(\delta k)}(p, q; M) = \frac{\mathcal{K}_0(p)}{p} \Theta(2-p) e^{-q^2/M^2}}. \quad (35)$$

There is no remaining integral. The p -shape is fixed; M acts only as the Gaussian width in q , so the GW spectrum peaks at $q \sim M$.

2. Decay model

For decaying turbulence, $\tilde{f}(\omega) = \hat{g}(\omega\tau_1)$ with $\hat{g}(\bar{q}) = e^{-i\bar{q}}(-i\bar{q})^{-1/3}\Gamma(\frac{1}{3}, -i\bar{q})$ and $\tau_1 = 1/\eta_0 = \sqrt{2\pi}/(Mk_0)$ the decay time at scale k_0 . With $q = \omega/k_0$ the natural temporal argument is $\bar{q} \equiv \omega\tau_1 = \sqrt{2\pi}q/M$; changing variables in Eq. (34), the temporal factor becomes $I_\omega = (1/\tau_1) \int dq_1 \hat{g}(q_1) \hat{g}(\bar{q} - q_1)$, which has no closed form. The dimensionless result is

$$\boxed{\mathfrak{H}_{\text{decay}}^{(\delta k)}(p, q; M) = \frac{\mathcal{K}_0(p)}{p} \Theta(2-p) \int_{-\infty}^{\infty} dq_1 \hat{g}(q_1) \hat{g}(\bar{q} - q_1), \quad \bar{q} = \frac{\sqrt{2\pi} q}{M}}. \quad (36)$$

The spatial part is fully resolved by the two δ -functions; only a 1-D frequency integral remains, and M enters it solely through the rescaled frequency $\bar{q}$.

B. Real-space delta-function correlations

Take the spatial part of R_{ij} to be a $\delta^{(3)}(\mathbf{r})$, with velocity variance v_0^2 and temporal decorrelation $T(\eta_k, \tau)$:

$$R_{ij}(\mathbf{r}, \tau) = v_0^2 \delta^{(3)}(\mathbf{r}) P_{ij} T(\eta_k, \tau), \quad \tilde{F}_{ij}(\mathbf{k}, \omega) = v_0^2 P_{ij}(\hat{k}) \tilde{T}(\eta_k, \omega), \quad (37)$$

so $A(k) = v_0^2$ is k -independent (i.e. $E(k) \propto k^2$, ultraviolet-dominated); the integrals must be cut off between k_0 and k_d . Crucially, we adopt the same k -dependent sweeping rate as in Sec. II, $\eta_k = \frac{1}{\sqrt{2\pi}} \varepsilon^{1/3} k^{2/3} = \frac{M}{\sqrt{2\pi}} k_0^{1/3} k^{2/3}$, so that $\tilde{T}$ depends on the wavenumber and the temporal factor no longer decouples from the spatial integral. With $A(k_1) = A(u) = v_0^2$ in Eq. (11) and dimensionless variables $z = k_1/k_0$, $y = u/k_0$, $p = k/k_0$, $q = \omega/k_0$, $R^{3/4} = k_d/k_0$,

$$H_{ijij}(\mathbf{k}, \omega) = \frac{v_0^4 k_0^2}{(2\pi)^7} \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{\mathcal{K}}(p, z, y) J(\omega; k_1, u), \quad J(\omega; k_1, u) \equiv \int d\omega_1 \tilde{T}(\eta_{k_1}, \omega_1) \tilde{T}(\eta_u, \omega - \omega_1), \quad (38)$$

where $\tilde{\mathcal{K}}(p, z, y) = \mathcal{K}(k, k_1, u)|_{k_1=z k_0, u=y k_0}$ and the frequency factor J now carries the k_1, u dependence inherited from η_{k_1}, η_u .

1. Kraichnan decorrelation

With $\tilde{T}(\eta_k, \omega) = (2/\eta_k) e^{-\omega^2/\pi\eta_k^2}$ the convolution combines two Gaussians of *different* widths η_{k_1}, η_u ; using $\int d\omega_1 e^{-a\omega_1^2} e^{-b(\omega-\omega_1)^2} = \sqrt{\pi/(a+b)} e^{-ab\omega^2/(a+b)}$ with $a = 1/\pi\eta_{k_1}^2$, $b = 1/\pi\eta_u^2$, and $\eta_{k_1}^2 + \eta_u^2 = \frac{M^2}{2\pi} k_0^{2/3} (k_1^{4/3} + u^{4/3})$, the frequency factor becomes

$$J(\omega; k_1, u) = \frac{4\pi\sqrt{2\pi}}{M k_0^{1/3} \sqrt{k_1^{4/3} + u^{4/3}}} \exp\left(-\frac{2\omega^2}{M^2 k_0^{2/3} (k_1^{4/3} + u^{4/3})}\right). \quad (39)$$

Inserting this into Eq. (38) and absorbing the constants into the dimensional prefactor gives the boxed dimensionless kernel

$$\mathfrak{H}_{\text{Kraichnan}}^{(\delta r)}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{K}(p, z, y) (z^{4/3} + y^{4/3})^{-1/2} e^{-2q^2/[M^2(z^{4/3} + y^{4/3})]}. \quad (40)$$

The exponential now acts as an M -dependent effective cutoff on the wavenumber integration: contributions require $z^{4/3} + y^{4/3} \gtrsim 2q^2/M^2$, so as M grows the effective wavenumber support widens and the GW spectrum peak shifts. This is precisely the structure of the Gogoberidze exponent $\exp[-2\varepsilon^{-2/3}\omega^2/(k_1^{4/3} + u^{4/3})]$ in Eq. (18); the k -dependent sweeping rate, rather than a fixed η_0 , is what makes M control the integration support.

2. Decay model

For decaying turbulence $\tilde{T}(\eta_k, \omega) = \hat{g}(\omega\tau_k)$ with the same $\hat{g}(q) = e^{-iq}(-iq)^{-1/3}\Gamma(\frac{1}{3}, -iq)$ as Sec. IV A 2, but now with the k -dependent decay time $\tau_k = 1/\eta_k = \sqrt{2\pi}/(Mk_0^{1/3}k^{2/3})$. The temporal factor is the convolution $J(\omega; k_1, u) = \int d\omega_1 \hat{g}(\omega_1\tau_{k_1})\hat{g}((\omega - \omega_1)\tau_u)$, which has no closed form and, because τ_{k_1}, τ_u depend on the wavenumbers, must remain *inside* the spatial integral. Writing $\tilde{J}(q; z, y, M)$ for this convolution with the per-mode dimensionless frequencies fixed by $\tau_{k_1}, \tau_u \propto 1/M$, the boxed result is

$$\mathfrak{H}_{\text{decay}}^{(\delta r)}(p, q; M, R) = \int_1^{R^{3/4}} \frac{z dz}{p} \int_{\max(|p-z|, 1)}^{\min(p+z, R^{3/4})} y dy \tilde{K}(p, z, y) \tilde{J}(q; z, y, M). \quad (41)$$

The decay rates scale as $\eta_k \propto Mk_0^{1/3}k^{2/3}$, so M again sets the effective wavenumber support. Unlike the fixed- η_0 case, the three integrals no longer factorize: the k -dependence of the decay rate couples the frequency convolution to the wavenumber integration, mirroring the behavior of Eq. (18).

Summary of the four dimensionless integrals

Spectrum	Decorrelation	Dimensionless integral
$E_0\delta(k - k_0)$	Kraichnan	Eq. (35): closed form; M in frequency scale only
$E_0\delta(k - k_0)$	Decay	Eq. (36): 1D frequency integral; M in $\bar{q}$
$\delta^{(3)}(\mathbf{r})$	Kraichnan	Eq. (40): 2D wavevector integral; M -cutoff in bounds
$\delta^{(3)}(\mathbf{r})$	Decay	Eq. (41): 2D wavevector $\times$ nested 1D frequency; M -cutoff

V. NUMERICAL RESULTS: $f_{\text{peak}}^{\text{GW}}(\tau_*, M_0)$

We scan the peak GW frequency $f_{\text{peak}}^{\text{GW}} = \text{argmax}_{\omega}[\omega \Omega_{\text{GW}}(\omega)]/2\pi$ over the (τ_*, M_0) plane for a delta-shell decaying source evolved with the BK2016 self-similar closure and time-integrated over the emission window (`Notebooks/f_peak_vs_tau_max`). Two temporal-correlation kernels are compared through their dimensionless transform $\mathcal{G}(q) = \widehat{R^2}(q)$: the random-phase BK2016 form (Model A) and the coherent eddy (Model B).

Across all four self-similar classes the ratio $f_{\text{peak}}^A/f_{\text{peak}}^B$ is *independent of* (τ_*, M_0) : the self-similar closure makes both the eddy frequency ω_e and the sweeping rate η_0 scale as $(1 + T/\tau_*)^{-1}$, so the two kernels inherit the same overall frequency scaling and differ only in their fixed dimensionless peak position. The model difference therefore collapses to a single per-class constant (Fig. 7): $f_A/f_B \simeq 0.37, 0.36$ for the HD (Loitsiansky, Saffman) classes and 0.83, 0.78 for the MHD (nonhelical, helical) classes – Model A always peaks lower. The large HD offset comes from the side-peak of the coherent kernel at the eddy frequency ($q \simeq \pm 2$, the $\cos \sigma$ oscillation of R_B), which wins the $\omega \Omega_{\text{GW}}$ argmax for the steep-decay HD classes but nearly coincides with the broad Model-A peak for MHD. Because the per-model maps differ only by this constant, we display the ratio rather than the eight individual heatmaps.

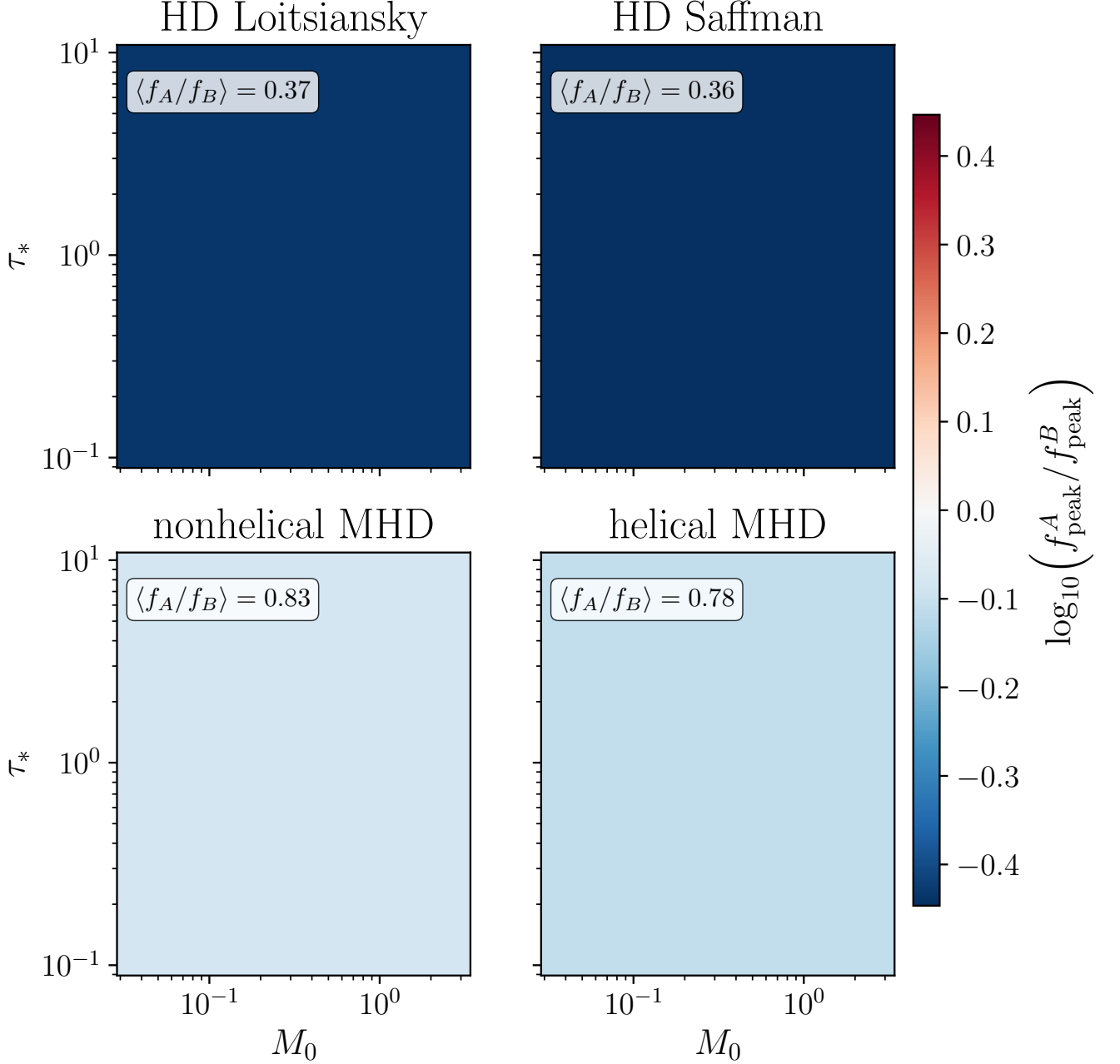


FIG. 7. Model A/B ratio of the peak GW frequency, $\log_{10}(f_{\text{peak}}^A/f_{\text{peak}}^B)$, on the (τ_*, M_0) plane for the four BK2016 self-similar decay classes (HD Loitsiansky $q = 2/7$, $\beta = 4$; HD Saffman $q = 1/3$, $\beta = 3$; nonhelical MHD $q = 1/2$, $\beta = 1$; helical MHD $q = 2/3$, $\beta = 0$). Each panel is flat: the ratio is independent of (τ_*, M_0) , fixed only by the kernel shapes, with the per-class value $\langle f_A/f_B \rangle$ annotated. Model A (random-phase) peaks below Model B (coherent eddy) everywhere; the HD classes show the larger offset.

VI. SPECTRUM SURVEY FOR ALL DERIVED MODELS

In this section we plot, on a common convention, the dimensionless GW energy-density spectrum

$$\Omega_{\text{GW}}(p) \propto p^3 H(p, p), \quad p = k_{\text{GW}}/k_0, \quad (42)$$

along the sound-cone diagonal $q = p$ for every model derived in this paper. The p^2 factor relative to the characteristic strain $h_c^2(p) = p H(p, p)$ is the phase-space measure d^3k ; together with causality and a finite white-noise

source it predicts $\Omega_{\text{GW}} \sim p^3$ in the IR, followed by a peak near $p \sim 1$ and a decay in the UV. A dotted grey line shows the p^3 reference; cases whose IR slope deviates from 3 by more than ± 0.5 are flagged in the build log of `Notebooks/all_spectra.py`. Models with a $1/p$ pole at the origin (monochromatic $E_0 \delta(k - k_0)$ source) cannot satisfy the universal scaling because the fluid itself carries no IR power.

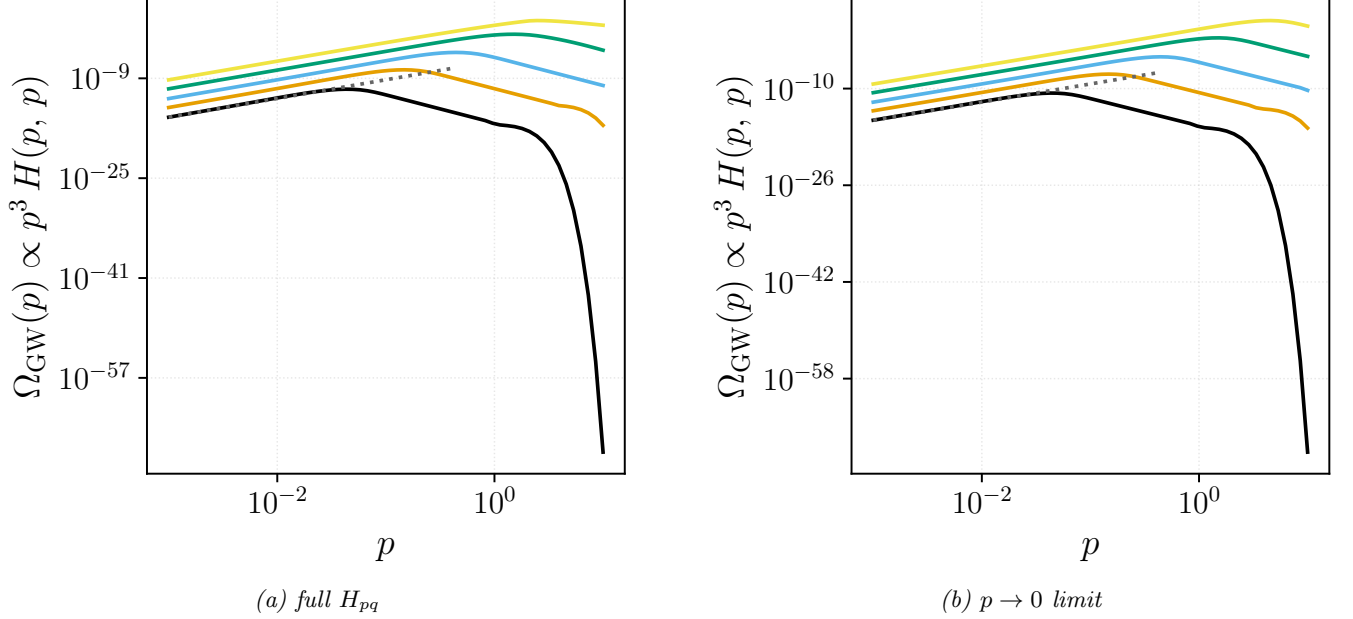


FIG. 8. Kraichnan decorrelation + Kolmogorov spectrum. Five curves per panel correspond to $M = 0.03, 0.1, 0.3, 1.0, 3.0$ at $R = 10^4$. Panel (a) is the full numerical H_{pq} (Sec. II); panel (b) is the aeroacoustic $p \rightarrow 0$ limit (Kahniashvili et al. 2008). Both give the universal IR slope $\Omega_{\text{GW}} \propto p^3$.

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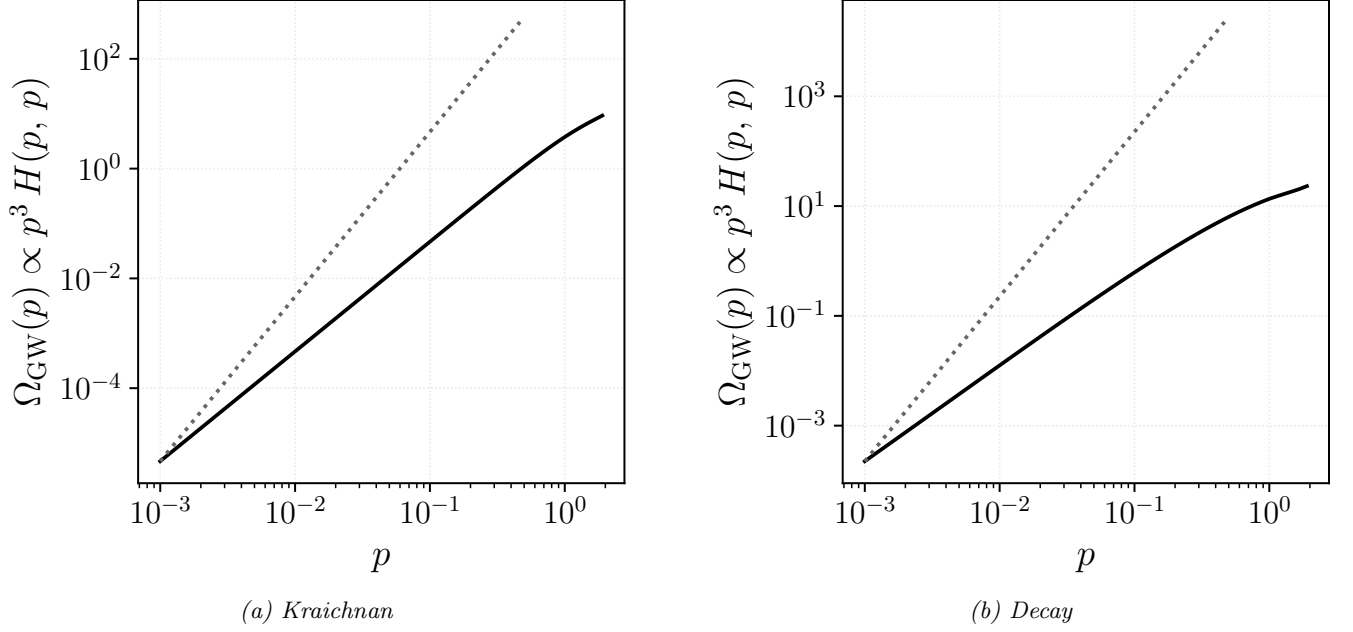


FIG. 9. Monochromatic spectrum $E(k) = E_0 \delta(k - k_0)$ (no free parameters at the dimensionless level). Panel (a): Kraichnan decorrelation, closed form $\Omega_{\text{GW}}(p) \propto p^3 K_0(p)/p \Theta(2-p) e^{-p^2/(2\pi)}$ with IR slope **2** (flagged: $1/p$ pole from the monochromatic source removes one power of p from the universal scaling). Panel (b): decay temporal convolution, IR slope **0.3** (flagged: the temporal-tail extra suppression further reduces the IR power).

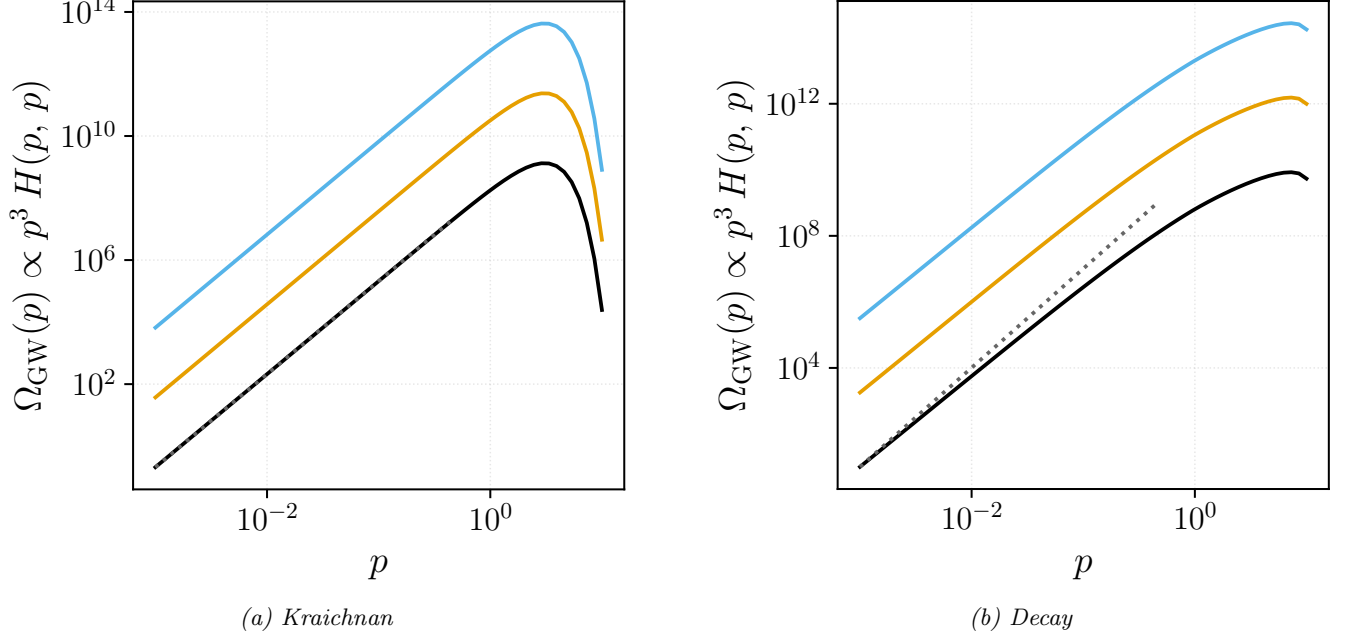


FIG. 10. White-noise real-space correlations $R_{ij} \propto \delta^{(3)}(\mathbf{r})$. Three curves per panel correspond to $R = 10^3, 10^4, 10^5$. Panel (a): Kraichnan decorrelation, matches the universal IR slope **3**. Panel (b): decay temporal model, IR slope **1.3** (flagged: temporal-convolution tails pull power down at small p).